

Skill 2:

```
#include <stdio.h>
#include <string.h>

#define MAX_INPUT 1024

int main() {

    char input[MAX_INPUT];

    printf("=====\n");
    printf("        Simple Linux Shell\n");
    printf("=====\n");

    while (1) {

        // Display prompt
        printf("mini-shell> ");
        fflush(stdout);

        // Read user input
        if (fgets(input, MAX_INPUT, stdin) == NULL) {
            printf("\nInput ended. Exiting shell...\n");
            break;
        }

        // Remove newline created by Enter key
        input[strcspn(input, "\n")] = '\0';

        // Ignore empty input
        if (strlen(input) == 0) {
            continue;
        }

        // Handle exit command
        if (strcmp(input, "exit") == 0) {
            printf("Exiting mini shell...\n");
            break;
        }

        // Display received command
        printf("You entered: %s\n", input);
    }

    return 0;
}
```

```

CC = gcc
CFLAGS = -Wall -Wextra

all: fork_demo exec_demo process_tree mini_shell

fork_demo:
    $(CC) $(CFLAGS) src/fork_demo.c -o bin/fork_demo

exec_demo:
    $(CC) $(CFLAGS) src/exec_demo.c -o bin/exec_demo

process_tree:
    $(CC) $(CFLAGS) src/process_tree.c -o bin/process_tree

mini_shell:
    $(CC) $(CFLAGS) src/mini_shell.c -o bin/mini_shell

clean:
    rm -f bin/*

```

```

nothing added to commit but untracked files present (use "git add" to track)
ritwik@DESKTOP-PK3LAMN:~$ cd ~/process_system_calls
ritwik@DESKTOP-PK3LAMN:~/process_system_calls$ tree
.
0 directories, 0 files
ritwik@DESKTOP-PK3LAMN:~/process_system_calls$ nano src/mini_shell.c
ritwik@DESKTOP-PK3LAMN:~/process_system_calls$ nano src/mini_shell.c
ritwik@DESKTOP-PK3LAMN:~/process_system_calls$ nano src/mini_shell.c
ritwik@DESKTOP-PK3LAMN:~/process_system_calls$ pwd
/home/bhanu/process_system_calls
ritwik@DESKTOP-PK3LAMN:~/process_system_calls$ ls
ritwik@DESKTOP-PK3LAMN:~/process_system_calls$ cd ~/process_system_calls
ritwik@DESKTOP-PK3LAMN:~/process_system_calls$ ls
ritwik@DESKTOP-PK3LAMN:~/process_system_calls$ mkdir -p src bin docs include
ritwik@DESKTOP-PK3LAMN:~/process_system_calls$ tree
.
├── bin
├── docs
├── include
└── src

5 directories, 0 files
ritwik@DESKTOP-PK3LAMN:~/process_system_calls$ nano src/mini_shell.c
ritwik@DESKTOP-PK3LAMN:~/process_system_calls$ nano Makefile
ritwik@DESKTOP-PK3LAMN:~/process_system_calls$ nano src/mini_shell.c
ritwik@DESKTOP-PK3LAMN:~/process_system_calls$ make mini_shell
gcc -Wall -Wextra src/mini_shell.c -o bin/mini_shell
ritwik@DESKTOP-PK3LAMN:~/process_system_calls$ ./bin/mini_shell
=====
Simple Linux Shell
=====
mini-shell> hello
You entered: hello
mini-shell> exit
Exiting mini shell...
ritwik@DESKTOP-PK3LAMN:~/process_system_calls$ |

```

```
> ^C
ritwik@DESKTOP-PK3LAMN1:~/process_system_calls$ fgets(input, MAX_INPUT, stdin)
-bash: syntax error near unexpected token `input,'
ritwik@DESKTOP-PK3LAMN1:~/process_system_calls$ ./bin/mini_shell
=====
      Simple Linux Shell
=====
mini-shell> Hello
You entered: Hello
mini-shell> hello world this is linux
You entered: hello world this is linux
mini-shell> abcd123
You entered: abcd123
mini-shell> exit
Exiting mini shell...
ritwik@DESKTOP-PK3LAMN1:~/process_system_calls$ |
```